Kernel Methods for the Computation of Quasi-Potentials in Stochastic Dynamical Systems

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Abstract

We present a data-driven, kernel-based framework for reconstructing the quasi-potential of stochastic differential equations (SDEs) in the weak-noise regime. The method approximates a family of Koopman eigenfunctions in a reproducing-kernel Hilbert space and combines them into a closed-form surrogate whose gradient automatically satisfies a quadratic Hamilton–Jacobi ansatz. A composite loss simultaneously (i) minimises the Hamilton–Jacobi residual on a collocation grid, (ii) matches the Hessian of the surrogate to the solution of the Riccati equation at the equilibrium, and (iii) promotes sparsity through an ℓ_1 penalty on the kernel-mixture weights.

Numerical experiments show that the analytic surrogate is inexpensive to evaluate, paving the way for efficient rare-event analysis and optimal-control applications in higher-dimensional, non-gradient stochastic systems.

1 Introduction

Noise-induced transitions and rare escape phenomena are ubiquitous in physical, chemical, biological, and engineering systems [31, 55]. The Freidlin-Wentzell large deviation theory serves as a fundamental framework for analyzing rare events driven by weak noise [10, 18, 33, 45, 46]. Central to this theory is the concept of the action functional, which quantifies the likelihood of trajectories deviating from deterministic paths. The quasi-potential, defined as the infimum of the action functional, lies at the heart of large deviation theory. It characterizes the probability of a stochastic system fluctuating into a neighborhood of a specific state and governs key quantities such as mean first exit times and stationary distributions.

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Despite its importance, computing the quasi-potential remains challenging. Traditional approaches can be categorized into three classes. Action Plot Methods [3]: These methods parameterize trajectories by integrating along unstable manifolds in the extended Hamiltonian phase space, starting from a small circle near a fixed point until exiting the basin of attraction. While effective for obtaining quasi-potential values along parameterized paths, they fail to establish a direct relationship between the quasi-potential and spatial coordinates. Geometric Minimum Action Method (GMAM) [30]: As a representative of string methods [12, 21, 52, 54], GMAM iteratively identifies the most probable path between a fixed point and a target state. However, it computes the quasi-potential only at discrete endpoints, lacking global structure. Ordered Upwind Method (OUM) [11, 17]: By discretizing the phase space and propagating quasi-potential values outward from fixed points, OUM achieves global quasi-potential reconstruction. Yet, its computational efficiency remains limited, especially in high-dimensional systems.

Recent advances in computational power and artificial intelligence have spurred the application of machine learning to stochastic dynamics. Techniques such as sparse regression [5, 39], Koopman operators [36, 53], and neural networks [14] have been leveraged for data-driven modeling of stochastic systems. In addition, some machine learning methods are also proposed to solve the problem of noise-induced transitions such as computing the most probable paths [40, 51] and the mean first exit times [42]. Notably, Lin et al. [43] proposed neural network-based frameworks to compute quasi-potentials for stochastic dynamical systems with additive Gaussian noise, while Li et al. [41] developed physics-informed neural networks (PINNs) to compute quasi-potentials by solving Hamilton-Jacobi equations.

Among machine learning paradigms, kernel methods stand out due to their strong nonlinear representation capabilities and computational efficiency. Notably, reproducing kernel Hilbert spaces (RKHS) [16] have established solid mathematical foundations for the analysis of dynamical systems [1, 4, 6–9, 19, 22–26, 28, 29, 32, 34–36, 38, 44] and surrogate modeling (cf. [47] for a survey) as well as analyzing neural networks [49]. The idea of combining multiple kernels by learning a weighted mixture is known as *multiple kernel learning* (MKL) [2, 37] and has been studied extensively in the machine learning literature. Our approach is related to MKL in that we learn a sparse convex combination of base kernels, but it differs in that the objective couples kernel selection with PDE residual minimization and curvature matching, rather than regression or classification loss.

In this work, we present a kernel-based algorithm for quasi-potential computation. The structure of this paper is as follows: Section 2 introduces the definition and properties of the quasi-potential. Section 3 details the proposed kernel-driven methodology. Numerical examples in Section 4 validate the algorithm's accuracy and efficiency, followed by concluding remarks.

2 Quasi-Potential and the Wentzel–Kramers–Brillouin Approximation in Stochastic Dynamical Systems

In this section, we examine dynamical systems governed by state variables $x_t \in \mathbb{R}^n$, evolving under the influence of Gaussian white noise with low intensity. The dynamics are described by the stochastic differential equation:

$$dx_t = b(x_t)\,dt + \sqrt{\varepsilon}\,\sigma(x_t)\,dB_t, \tag{1}$$

where $B_t \in \mathbb{R}^n$ denotes an n-dimensional standard Brownian motion, and ε is a small positive parameter quantifying the noise strength. The function $b(x) \in \mathbb{R}^n$ represents the

deterministic drift term, which we assume has continuously differentiable derivatives. The diffusion coefficient is given by the matrix $a(x) = \sigma(x)\sigma(x)^\top$, where we assume $a(x)$ is uniformly positive definite on Ω (ensuring non-degeneracy of the noise).

We suppose $\bar{x}$ is an **asymptotically stable and hyperbolic** fixed point of the associated deterministic system, that is, $b(\bar{x}) = 0$ and all eigenvalues of its linearization matrix $J = \nabla b(\bar{x})$ have **strictly negative real parts**, i.e., $\mathrm{Re}(\lambda_i) < 0$ for all $i = 1, \ldots, n$. This ensures asymptotic stability in the sense that trajectories starting near $\bar{x}$ converge to $\bar{x}$ exponentially in time. Under stochastic perturbations, the process x_t explores the state space and converges to a stationary probability distribution $p_s(x)$. In the regime of weak noise ($\varepsilon \ll 1$), this stationary distribution admits a Wentzel–Kramers–Brillouin (WKB)-type asymptotic approximation (see [15, 41, 48]):

$$p_s(x) \sim C(x) \exp\left\{-\varepsilon^{-1} V(x)\right\}, \tag{2}$$

where $C(x)$ is a slowly varying prefactor and $V(x)$ is the quasi-potential function, characterizing the exponential decay rate of the probability density. This form is termed a *WKB-type approximation* named after the WKB method in quantum mechanics, where solutions in the semiclassical limit $\hbar \to 0$ take the form $\psi(x) \sim A(x) \exp\{iS(x)/\hbar\}$. In both contexts, the exponential term dominates and captures the leading-order behavior, while the prefactor varies more slowly. In stochastic systems, the small parameter ε appears in the exponent, leading to sharp concentration of the probability distribution around stable points as $\varepsilon \to 0$. The quasi-potential $V(x)$ acts as an effective potential or energy landscape, quantifying the cost of rare fluctuations. The symbol "$\sim$" denotes logarithmic equivalence, meaning that

$$\lim_{\varepsilon \to 0} \frac{\log p_s(x)}{-\varepsilon^{-1} V(x)} = 1,$$

which highlights that $V(x)$ governs the leading-order behavior in the small noise limit.

The concept of the quasi-potential originates from large deviation theory [46]. It is defined via a variational principle involving the Freidlin–Wentzell action functional $S_T[\phi]$, formulated over continuous trajectories $\phi(t)$ that connect a stable fixed point $\bar{x}$ to a target point x. Specifically, the quasi-potential is given by the nested infimum:

$$V(x) \equiv \inf_{T>0} \inf_{\phi \in C([0,T])} \left\{ S_T[\phi] \,:\, \phi(0) = \bar{x},\ \phi(T) = x \right\}, \tag{3}$$

where $C([0,T])$ denotes the space of continuous functions on the interval $[0,T]$. The action functional $S_T[\phi]$, fundamental in the large deviation framework, is given by

$$S_T[\phi] = \int_0^T L(\phi(t), \dot{\phi}(t))\, dt, \tag{4}$$

with the Lagrangian $L(\phi, \dot{\phi})$ for the stochastic differential equation (1) defined as follows

$$L(\phi, \dot{\phi}) = \frac{1}{2} \left[\dot{\phi} - b(\phi)\right]^T a^{-1}(\phi) \left[\dot{\phi} - b(\phi)\right]. \tag{5}$$

This action functional quantifies the cost of deviations from the deterministic path and plays a central role in describing the asymptotic probability of rare events. In particular, for a fixed trajectory $\phi(t)$, the probability that the stochastic process x_t remains within a small tube around $\phi(t)$ over the time interval $[0,T]$ satisfies the large deviation estimate:

$$\mathbb{P}_{x_0}\left\{ \sup_{t \in [0,T]} |x_t - \phi(t)| < \delta \right\} \sim \exp\left\{-\varepsilon^{-1} S_T[\phi]\right\}, \tag{6}$$

for small $\delta > 0$, illustrating that the action functional governs the exponential rate of decay of such pathwise probabilities.

Note that the stationary probability distribution satisfies the time-independent Fokker–Planck equation

$$\mathcal{L}p_s(x) = -\sum_{i=1}^{n} \frac{\partial}{\partial x_i} b_i(x) p_s(x) + \sum_{i,j=1}^{n} \frac{\partial^2}{\partial x_i \partial x_j} a_{ij}(x) p_s(x) = 0,$$

where $\mathcal{L}$ is the Fokker–Planck operator. Substituting the WKB ansatz (2) into this equation and collecting the leading-order terms in ε yields the Hamilton–Jacobi equation:

$$H(x,p) \equiv b^T p + \frac{1}{2} p^T a p = 0, \quad p \equiv \nabla V(x), \tag{7}$$

which is a first-order nonlinear partial differential equation for the quasi-potential V. It is observed that this Hamiltonian is the Legendre transform of the Lagrangian (5). According to the definition (3), the quasi-potential at the fixed point satisfies $V(\bar{x}) = 0$. Since the quasi-potential function attains minimal value at the stable fixed point, the gradient of the quasi-potential is also zero at the stable point, i.e., $\nabla V(\bar{x}) = 0$.

If $b(x) = -\nabla U(x)$ (i.e., a gradient system) and $a(x) = I$ (i.e., additive noise), then the quasi-potential function equals twice the potential: $V(x) = 2U(x)$. In this special case, the quasi-potential admits an analytical expression.

Assume that the second-order approximation of the quasi-potential function at the fixed point $\bar{x}$ is given by

$$\begin{aligned} V(x) &\approx V(\bar{x}) + \nabla V(\bar{x})(x-\bar{x}) + \frac{1}{2}(x-\bar{x})^T \nabla^2 V(\bar{x})(x-\bar{x}) \\ &= \frac{1}{2}(x-\bar{x})^T Z (x-\bar{x}), \end{aligned}$$

where $Z = \nabla^2 V(\bar{x})$. Consequently, its gradient is approximated as

$$\nabla V(x) \approx Z(x-\bar{x}).$$

Substituting the gradient approximation into the Hamilton–Jacobi equation (7) yields

$$b(x)^T Z(x-\bar{x}) + \frac{1}{2}(x-\bar{x})^T Z^T a(x) Z (x-\bar{x}) \approx 0. \tag{8}$$

Near the fixed point, we also approximate the drift $b(x)$ and the diffusion $a(x)$ as

$$b(x) \approx F(x-\bar{x}), \ a(x) \approx A,$$

where $F \equiv \nabla b(\bar{x}), A \equiv a(\bar{x})$. Substituting the linearized $b(x)$ and $a(x)$ into Eq. (8), we have

$$(x-\bar{x})^T (F^T Z + \frac{1}{2} Z^T A Z)(x-\bar{x}) \approx 0. \tag{9}$$

Note that it is equivalent to the following equation.

$$(x-\bar{x})^T (F^T Z + ZF + Z^T A Z)(x-\bar{x}) \approx 0. \tag{10}$$

Since the matrix $F^T Z + ZF + Z^T A Z$ is a symmetric matrix, it must be diagonalizable. Due to the arbitrariness of $x - \bar{x}$, the eigenvalues of the matrix must all be 0. That is,

each element of the matrix is 0. Therefore, the Hessian matrix $Z = \nabla^2 V(\bar{x})$ satisfies the algebraic Riccati equation:

$$F^T Z + ZF + Z^T A Z = 0 \tag{11}$$

or the equivalent form:

$$FZ^{-1} + Z^{-1}F^T + A = 0. \tag{12}$$

The principal challenge in computing the quasi-potential from the PDE (7) lies in avoiding trivial solutions. To address this, we employ kernel methods to solve the PDE (7), and use equation (12) to impose that the Hessian of V at the origin coincides with the matrix Z solving (12).

It is worth emphasizing that the quasi-potential plays a pivotal role not only in large deviation theory but also in stochastic dynamics, as it is linked to various physical quantities. For example, the expected time to exit the domain of attraction of a fixed point—known as the mean exit time—is exponentially determined by the minimum of the quasi-potential on the boundary of the domain. The exit location distribution and the invariant measure can also be approximated based on the quasi-potential. Furthermore, once the quasi-potential function is known, the most probable transition path connecting two steady states can be derived from the quasi-potential. Typically, the most probable path passes through a saddle point, with the latter half of the trajectory beyond the saddle point following its unstable manifold toward the destination steady state. The first half of the path is obtained by solving the equation

$$\dot{x} = b(x) + a(x)\nabla V(x). \tag{13}$$

The boundary conditions are $x(-\infty) = \bar{x}$ and $x(\infty) = x_{sa}$, where x_{sa} denotes the saddle point. In practice, Equation (13) is obtained by substituting $p = \nabla V(x)$ into the state equation $\dot{x} = b(x) + a(x)p$ of the Hamiltonian system corresponding to the Hamilton-Jacobi Equation (7). Therefore, the most probable path is the projection of a heteroclinic orbit in the four-dimensional Hamiltonian phase space onto the two-dimensional space. For further properties of the quasi-potential, see [46].

Remark 2.1. When $\varepsilon = 0$ (i.e. when there is no noise), our work will reduce to constructing the Lyapunov function for the unforced system $\dot{x} = b(x)$ [20, 38]

3 Method for Approximating the Quasi-Potential

To approximate the quasi-potential, we exploit the spectral structure of the Koopman operator. Specifically, we construct eigenfunctions $\varphi_i(x)$ satisfying the linear advection-type partial differential equations:

$$\nabla\varphi_i(x) \cdot b(x) = \lambda_i \varphi_i(x), \quad \varphi_i(\bar{x}) = 0, \quad \nabla\varphi_i(\bar{x}) = e_i, \tag{14}$$

where λ_i are eigenvalues of the linearization $J = \nabla b(\bar{x})$ at the fixed point $\bar{x}$, and e_i are the standard basis vectors. These eigenfunctions correspond to the Koopman eigenfunctions associated with the equilibrium dynamics and encode local geometric and dynamical information such as the structure of stable and unstable manifolds.

Let $\Phi(x) = (\varphi_1(x) \cdots \varphi_n(x))^T$, and define the matrix W whose columns are the eigenvectors w_i of J, satisfying $Jw_i = \lambda_i w_i$. We then define a quadratic approximation to the quasi-potential:

$$V_{\text{mid}}(x) = \tfrac{1}{2}\Phi(x)^T P_{\text{mid}} \Phi(x), \tag{15}$$

with the symmetric matrix[1]:

$$P_{\text{mid}} = W \,\text{diag}(-2\lambda_i)\, W^{-1}. \tag{16}$$

To represent $\varphi_i(x)$, we employ a data-driven kernel-based ansatz:

$$\varphi_i(x) = \sum_{p=1}^{N} \alpha_p^{(i)}\, k_\beta(x, x_p) + (x - \bar{x}) \cdot w_i, \tag{17}$$

where the point set $\{x_p\}_{p=1}^N$ denotes the collocation set which is used for training, and $k_\beta(x,t)$ is a convex combination of base kernels (e.g., polynomials and radial basis functions), parameterized by a non-negative weight vector β. This kernel formulation ensures flexibility and regularization, and it circumvents the need to construct the Koopman operator explicitly.

The coefficients α_p are determined by minimizing the regularized empirical loss functional using the representer theorem. To determine the coefficients $\alpha^{(i)} = (\alpha_1^{(i)}, \ldots, \alpha_N^{(i)})$, we enforce the Koopman eigenfunction equation at the collocation points. Specifically, for each $i = 1, \ldots, n$, we impose the conditions

$$\varphi_i(\bar{x}) = 0, \qquad \nabla\varphi_i(\bar{x}) = e_i,$$

together with the PDE constraint

$$\nabla\varphi_i(x_p) \cdot b(x_p) = \lambda_i\, \varphi_i(x_p), \qquad p = 1, \ldots, N.$$

Substituting the RKHS ansatz (17) into these conditions yields a linear system for the coefficients $\alpha^{(i)}$, which we solve in a regularized least–squares sense:

$$\min_{\alpha^{(i)}} \sum_{p=1}^{N} \left(\nabla\varphi_i(x_p) \cdot b(x_p) - \lambda_i\, \varphi_i(x_p)\right)^2 \;+\; \eta\, \|\varphi_i\|_{\mathcal{H}_k}^2, \qquad \|\varphi_i\|_{\mathcal{H}_k}^2 = \alpha^{(i)T} G_\beta \alpha^{(i)}. \tag{18}$$

Here $(G_\beta)_{pq} = k_\beta(x_p, x_q)$ is the kernel Gram matrix. The solution of this regularized linear system provides the coefficients $\alpha^{(i)}$ used to construct $\Phi(x)$ and the quasi-potential surrogate $V_{\text{mid}}(x)$.

Remark 3.1 (RKHS norm and the linear correction term)**.** The ansatz (17) decomposes φ_i into an RKHS part $\sum_p \alpha_p^{(i)} k_\beta(x, x_p)$ and a fixed linear term $(x - \bar{x}) \cdot w_i$. The RKHS norm $\|\varphi_i\|_{\mathcal{H}_k}^2 = \alpha^{(i)T} G_\beta \alpha^{(i)}$ penalizes only the learnable kernel component, not the linear correction. This is intentional: the linear term is analytically prescribed by the boundary condition $\nabla\varphi_i(\bar{x}) = e_i$ and carries zero free parameters, so it should not be regularized. Penalizing it would bias the solution away from the exact local linearization.

To determine optimal kernel weights β, we minimize the following regularized loss function, where $P_{\text{loc}} \in \mathbb{R}^{n\times n}$ denotes the local curvature target matrix—specifically, the positive-definite solution of the algebraic Riccati equation (12) evaluated at $\bar{x}$:

[1] Alternatively, one can set $P_{\text{mid}} = \frac{1}{2}(P_u + P_\ell)$, where P_u and P_ℓ solve Riccati-type equations: $P\Lambda + \Lambda^\top P + PRP = 0$, for different choices of R.

$$L(\beta) = \underbrace{\sum_p \left[b^T(x_p)\nabla V_{\mathrm{mid}}(x_p) + \frac{1}{2}\nabla V_{\mathrm{mid}}(x_p)^T a(x_p)\nabla V_{\mathrm{mid}}(x_p) \right]^2}_{L_{\mathrm{HJ}}} + \gamma \underbrace{\|\nabla^2 V_{\mathrm{mid}}(\bar{x}) - P_{\mathrm{loc}}\|_F^2}_{L_{\mathrm{curv}}}$$
$$+ \lambda_1 \sum_j |\beta_j| + \eta \sum_i \|\varphi_i\|_{\mathcal{H}_k}^2, \tag{19}$$

subject to the constraints $\beta_j \geq 0$ and $\sum_j \beta_j = 1$. Here $\|\cdot\|_F$ denotes the Frobenius norm. The first term L_{HJ} enforces consistency with the Hamilton–Jacobi structure; the second term L_{curv} aligns the local curvature of the approximation with that of the Riccati solution at $\bar{x}$. The remaining terms promote sparsity and RKHS regularity.

Algorithmic structure. In practice, the computation proceeds in two stages. For a fixed choice of kernel weights β, the coefficients $\alpha^{(i)}$ are obtained by solving the regularised linear systems (18), yielding the Koopman eigenfunction approximations φ_i and hence $\Phi(x)$. In a second, outer optimisation loop, the kernel mixture coefficients β are updated by minimising the composite loss (19), while at each iteration the functions φ_i are recomputed from the updated kernel k_β. This bi-level structure (inner linear solves for $\alpha^{(i)}$, outer nonlinear optimisation for β) matches the implementation used in the numerical experiments of Section 4.

Kernel dictionary. Throughout our numerical experiments, we employ a hybrid dictionary combining Gaussian radial basis function (RBF) kernels and polynomial kernels. This choice is motivated by complementary approximation properties: RBF kernels excel at local interpolation and capturing rapid variations near critical points, while polynomial kernels naturally encode global algebraic structure characteristic of quasi-potentials. In our previous work [27], this RBF+polynomial combination demonstrated robust performance across diverse dynamical systems benchmarks. The ℓ_1 sparsity penalty in the composite loss (19) acts as an automatic model selection mechanism analogous to MKL [2, 37]: kernels that do not contribute meaningfully to the approximation have their weights driven to zero during optimization. In principle, the framework can accommodate much larger and more diverse kernel dictionaries; in [44] the same ℓ_1-based sparsification mechanism was used with an extensive kernel library to learn 132 chaotic dynamical systems. The methodology naturally extends to other kernel families (e.g., Matérn, rational quadratic, or domain-specific kernels) when more complex functional structures are anticipated.

Theoretical guarantee. The composite loss (19) couples three ingredients:

i. a Hamilton–Jacobi residual that enforces the global large-deviation structure,

ii. an ℓ_1 / RKHS regulariser that controls capacity, and

iii. a Riccati–curvature term that anchors the surrogate to the exact quadratic well at the attractor.

The next result shows that, under the standard assumptions of kernel-ridge learning and an appropriate schedule for the regularisation weights, these ingredients are sufficient to drive the surrogate V_N to the true quasi-potential V as the number of collocation points N

grows. In particular, the Hessian of V_N at the equilibrium converges to the Riccati matrix P, giving a second-order match that accelerates the finite-sample error decay observed in Section 4. We state the result and the proof for completeness.

Theorem 3.2 (Consistency with Riccati–curvature matching)**.** *Let $V : \Omega \subset \mathbb{R}^d \to \mathbb{R}_{\geq 0}$ be the unique viscosity solution of the stationary Hamilton–Jacobi equation*

$$H\big(x, \nabla V(x)\big) = 0, \qquad V(\bar{x}) = 0,$$

and let $P \succ 0$ be the Riccati matrix

$$b'(\bar{x})^\top P + P\, b'(\bar{x}) \;=\; P\, a(\bar{x})\, P.$$

Assume:

1. *$b, a \in C^1(\Omega)$ and $a(x)$ is uniformly positive definite on Ω;*
2. *k is a continuous, C^m $(m \geq 2)$* universal *kernel on Ω with RKHS $\mathcal{H}_k$;*
3. *Collocation points $\{x_p\}_{p=1}^N$ are drawn i.i.d. from a probability measure ρ whose support is $\overline{\Omega}$;*
4. *Regularisation parameters follow $\lambda_N \downarrow 0$, $N\lambda_N \uparrow \infty$, and a curvature weight γ_N satisfies $\gamma_N \downarrow 0$ with $\gamma_N/\lambda_N = O(1)$.*

For $N \in \mathbb{N}$, define the empirical objective:

$$\mathcal{L}_{N,\lambda,\gamma}(\Phi, \beta) := \frac{1}{N}\sum_{p=1}^{N} H\big(x_p, \nabla V_{(\Phi,\beta)}(x_p)\big)^2 \;+\; \gamma \big\| D^2 V_{(\Phi,\beta)}(\bar{x}) - P \big\|_F^2 \;+\; \lambda\big(\|\Phi\|_{\mathcal{H}_k}^2 + \|\beta\|_1\big), \tag{20}$$

and let $(\widehat{\Phi}_N, \widehat{\beta}_N)$ be any minimiser of (20). Set $V_N := V_{(\widehat{\Phi}_N, \widehat{\beta}_N)}$.

Then, as $N \to \infty$,

$$\|V_N - V\|_{L^2(\rho)} \xrightarrow{p} 0, \qquad \|\nabla V_N - \nabla V\|_{L^2(\rho)} \xrightarrow{p} 0, \qquad D^2 V_N(\bar{x}) \xrightarrow{p} P.$$

If $V \in C^m(\Omega)$, the convergence is uniform on compact subsets of Ω.

Remark 3.3 (Notation)**.** In the theoretical analysis (Theorem 3.2), we use V_N to denote the N-sample approximation obtained by minimizing the empirical objective. In the numerical examples (Section 4), we use V_{mid} to denote the mid-level Koopman-based approximation constructed via Equation (15), and V_h or V_{quad} for specific baseline or hybrid approximations. These notational variants all refer to finite-sample approximations of the true quasi-potential V, distinguished only by their construction method or the context in which they appear.

Proof of Theorem 3.2. We follow the classical excess–risk decomposition of kernel ridge regression [13, 50], extended with curvature matching.

Step 1: Generalized representer theorem. The empirical loss in (20) depends on three types of evaluations: function values and gradients at the collocation points, and second derivatives at a single fixed point $\bar{x}$. By the generalized representer theorem, this structure implies that V_N admits a finite-dimensional representation:

$$V_N(x) = \sum_{p=1}^{N}\sum_{j=1}^{n} c_{pj}\,\partial_{x_j}k(x,x_p) + \sum_{1\le j\le \ell\le n} d_{j\ell}\,\partial_{x_j x_\ell}k(x,\bar{x}),$$

with coefficients $\{c_{pj}\},\{d_{j\ell}\}$ determined by the optimiser. All derivatives belong to $\mathcal{H}_k$ by the C^m assumption on k, and the RKHS norm is bounded by

$$\|V_N\|_{\mathcal{H}_k} \;\le\; C\|\theta_N\|_2,$$

where θ_N stacks all the coefficients.

Step 2: Population risk and curvature penalty. Define the population risk:

$$R(V) \;:=\; \mathbb{E}_\rho\big[H\big(X,\nabla V(X)\big)^2\big], \qquad C(V) \;:=\; \big\|D^2V(\bar{x}) - P\big\|_F^2,$$

and the regularised functional:

$$\mathcal{J}_{\lambda,\gamma}(V) := R(V) + \gamma C(V) + \lambda\|V\|_{\mathcal{H}_k}^2.$$

Since $a(x) \succ 0$, the Hamiltonian H is strictly convex in its second argument, ensuring that V is the unique viscosity solution and also the unique minimiser of $\mathcal{J}_{\lambda,\gamma}$ (as $C(V)=0$ and the regularisers vanish at V).

Step 3: Empirical risk decomposition. Define the empirical risk:

$$\widehat{R}_N(V) := \frac{1}{N}\sum_{p=1}^{N} H\big(x_p,\nabla V(x_p)\big)^2.$$

Then V_N minimises the regularised empirical objective:

$$\widehat{\mathcal{J}}_N(V) := \widehat{R}_N(V) + \gamma_N C(V) + \lambda_N\|V\|_{\mathcal{H}_k}^2.$$

By standard excess risk decomposition:

$$R(V_N) - R(V) = \big[R(V_N) - \widehat{R}_N(V_N)\big] + \big[\widehat{R}_N(V_N) - \widehat{R}_N(V)\big] + \big[\widehat{R}_N(V) - R(V)\big].$$

The middle term is $\le -\gamma_N C(V_N) - \lambda_N(\|V_N\|_{\mathcal{H}_k}^2 - \|V\|_{\mathcal{H}_k}^2)$ by optimality of V_N. Hence,

$$R(V_N) - R(V) + \gamma_N C(V_N) \;\le\; 2\sup_{V\in\mathcal{B}_N}\big|R(V) - \widehat{R}_N(V)\big|, \tag{21}$$

Define the comparison ball

$$\mathcal{B}_N := \Big\{f\in\mathcal{H}_k : \|f\|_{\mathcal{H}_k} \le \|V\|_{\mathcal{H}_k} + \lambda_N^{-1/2}\Big\},$$

where V denotes the true quasi-potential (as in the Theorem statement), so that $V_N \in \mathcal{B}_N$ with high probability.

Step 4: Uniform deviation bound. Since k is bounded and smooth on compact Ω, and $H(x,\cdot)$ is Lipschitz in $\nabla V(x)$, the Lipschitz constant over $\mathcal{B}_N$ is $O(\lambda_N^{-1/2})$. Talagrand–Bousquet inequality gives:

$$\sup_{V\in\mathcal{B}_N}\big|R(V) - \widehat{R}_N(V)\big| \le C\left(\sqrt{\frac{\log(1/\delta)}{N}}\,\lambda_N^{-1/2} + \frac{\log(1/\delta)}{N}\right)$$

with high probability. Therefore,

$$\sup_{V \in \mathcal{B}_N} \left| R(V) - \widehat{R}_N(V) \right| = O_p\left(\sqrt{\frac{1}{N\lambda_N}} \right).$$

Here, the notation $\mathcal{O}_p(\epsilon_N)$ means that the random variable is bounded in probability at rate ϵ_N, i.e., for all $\delta > 0$ there exists $M > 0$ such that $\mathbb{P}(|Z_N| > M\epsilon_N) < \delta$ for all sufficiently large N.

Step 5: Convergence in risk and curvature. The assumption $N\lambda_N \to \infty$ implies that the RHS of (21) vanishes in probability. Since $\gamma_N/\lambda_N = O(1)$ and $\gamma_N \to 0$, it follows that:

$$R(V_N) - R(V) \to 0, \qquad C(V_N) \to 0 \quad \text{in probability.}$$

Step 6: Convergence in L^2 and gradient.
The coercivity of $a(x)$ yields:

$$\tfrac{1}{2}\lambda_{\min}(a)\, \|\nabla V_N - \nabla V\|^2_{L^2(\rho)} \;\le\; R(V_N) - R(V),$$

hence $\nabla V_N \to \nabla V$ in $L^2(\rho)$, and $V_N \to V$ in $L^2(\rho)$ by Poincaré–Friedrichs on bounded Ω.

Step 7: Hessian consistency. $C(V_N) \to 0$ implies

$$D^2 V_N(\bar{x}) \to P \quad \text{in Frobenius norm, and hence in probability.}$$

Step 8: Uniform convergence (optional). If $V \in C^m(\Omega)$ and $k \in C^m$, the H_k norm controls m derivatives. By Sobolev embedding or Arzelà–Ascoli, convergence is uniform on compact subsets:

$$V_N \to V \quad \text{in } C^{m-1}(K) \quad \text{for all compact } K \Subset \Omega,$$

where $K \Subset \Omega$ implies that K is a compact subset whose closure lies strictly inside Ω. This ensures that the convergence statements avoid potential boundary irregularities, which is standard in the analysis of convergence in function spaces. □

Theorem 3.2 establishes that the learned quasi-potential V_N converges to the true solution V of the Hamilton–Jacobi equation in both value and gradient. In particular, the penalty term $\|D^2 V_N(\bar{x}) - P\|_F^2$ enforces curvature consistency at the stable equilibrium $\bar{x}$. To clarify the necessity and interpretation of this Riccati-based curvature constraint, we include the following corollary, which shows that any sufficiently regular solution of the Hamilton–Jacobi equation must satisfy this Riccati condition locally at $\bar{x}$.

Corollary 3.4 (Local Consistency with the Hamilton–Jacobi Equation)**.** *Let $V \in C^2(\Omega)$ solve*

$$H(x, \nabla V(x)) = f(x)^\top \nabla V(x) + \tfrac{1}{2}\nabla V(x)^\top a(x)\, \nabla V(x) = 0, \qquad V(\bar{x}) = 0,$$

with $\bar{x}$ a globally asymptotically stable equilibrium of f. Then the Hessian of V at $\bar{x}$ equals the Riccati matrix P:

$$D^2 V(\bar{x}) = P,$$

where P is the unique positive-definite solution to the algebraic Riccati equation

$$b'(\bar{x})^\top P + P\, b'(\bar{x}) = P\, a(\bar{x})\, P.$$

Proof. Near the equilibrium point $\bar{x}$, write the Taylor expansion

$$V(x) = \tfrac{1}{2}(x-\bar{x})^\top Z(x-\bar{x}) + o\big(\|x-\bar{x}\|^2\big), \qquad Z = D^2V(\bar{x}).$$

Linearise the drift and diffusion at $\bar{x}$:

$$f(x) = b'(\bar{x})(x-\bar{x}) + o(\|x-\bar{x}\|), \qquad a(x) = a(\bar{x}) + o(1).$$

Thus the gradient satisfies $\nabla V(x) = Z(x-\bar{x}) + o(\|x-\bar{x}\|)$. Insert these expansions into the Hamilton–Jacobi equation:

$$H(x, \nabla V(x)) = f(x)^\top \nabla V(x) + \tfrac{1}{2}\nabla V(x)^\top a(x)\, \nabla V(x) = 0.$$

Collecting second-order terms in $x - \bar{x}$ yields

$$(x-\bar{x})^\top \big(b'(\bar{x})^\top Z + Z b'(\bar{x}) + Z a(\bar{x}) Z\big)(x-\bar{x}) + o\big(\|x-\bar{x}\|^2\big) = 0.$$

Since this holds for all x near $\bar{x}$, the symmetric coefficient must vanish:

$$b'(\bar{x})^\top Z + Z b'(\bar{x}) + Z a(\bar{x}) Z = 0.$$

By standard results on algebraic Riccati equations (positive-definite $a(\bar{x})$ and stability of $b'(\bar{x})$), there is a unique positive-definite solution P of this equation. Hence $Z = P$ and therefore $D^2V(\bar{x}) = P$. □

4 Examples

4.1 Example 1: The two-Dimensional Maier–Stein System

We consider the Maier–Stein system whose deterministic part is

$$\dot{x} = F(x) = \begin{pmatrix} x_1 - x_1^3 - x_1 x_2^2 \\ -(1 + x_1^2) x_2 \end{pmatrix},$$

which features a stable fixed point at $\bar{x} = (-1, 0)$. Throughout this example, we assume the diffusion matrix is the identity, i.e., $a(x) \equiv I$. Note that the corresponding deterministic system is a gradient system with the potential

$$U(x) = \frac{1}{4}(x_1^2 - 1)^2 + \frac{1}{2}(x_1^2 + 1)x_2^2.$$

The associated quasi-potential in this case is given by (here and throughout the examples, V_{true} denotes the analytically known quasi-potential, i.e., the true solution V from Theorem 3.2)

$$V_{\text{true}}(x) = 2U(x) = \frac{1}{2}(x_1^2 - 1)^2 + (x_1^2 + 1)x_2^2.$$

Linearization at the fixed point gives the Jacobian

$$J = DF(\bar{x}), \quad Jw_i = \lambda_i w_i,$$

where (λ_i, w_i) are the eigenpairs of J. We solve the Lyapunov equation

$$JZ + ZJ^\top = -I, \quad \Rightarrow \quad P_{\text{loc}} = Z^{-1},$$

yielding $P_{\text{loc}} = \nabla^2 V(\bar{x})$, the Hessian of the quasi-potential at the saddle.

We seek eigenfunctions $\varphi_i(x)$ satisfying the Koopman PDE:

$$\nabla\varphi_i(x) \cdot F(x) = \lambda_i \varphi_i(x), \quad \varphi_i(\bar{x}) = 0, \quad \nabla\varphi_i(\bar{x}) = e_i.$$

Each φ_i is represented in an RKHS:

$$\varphi_i(x) = \sum_{p=1}^{N} \alpha_p^{(i)} k_\beta(x, x_p) + (x - \bar{x}) \cdot w_i, \quad k_\beta(x,t) = \sum_{j=1}^{L} \beta_j k_j(x,t).$$

On a collocation set $\{x_p\}_{p=1}^N$, we impose the constraints:

$$\begin{aligned} &\varphi_i(\bar{x}) = 0, \\ &\nabla\varphi_i(\bar{x}) \cdot e_j = \delta_{ij}, \quad j = 1, 2, \\ &\nabla\varphi_i(x_p) \cdot F(x_p) = \lambda_i (x_p - \bar{x}) \cdot w_i, \quad p = 1, \ldots, N. \end{aligned}$$

Solving the regularized linear system yields the coefficients $\alpha^{(i)}$.

Define $\Phi(x) = (\varphi_1(x), \varphi_2(x))^\top$, and construct the quasi-potential:

$$V_{\text{mid}}(x) = \tfrac{1}{2}\, \Phi(x)^T P_{\text{mid}} \Phi(x)$$

with

$$P_{\text{mid}} = W\, \text{diag}(-2\lambda_i)\, W^{-1}.$$

To determine β, we minimize the composite loss (19) (the same L_{HJ}, L_{curv} terms defined in Section 3):

$$L(\beta) = L_{\text{HJ}} + \gamma L_{\text{curv}} + \lambda_1 \sum_{j=1}^{L} |\beta_j| + \eta \sum_{i=1}^{2} \|\varphi_i\|_{\mathcal{H}_k}^2,$$

where

$$L_{\text{HJ}} = \sum_{p=1}^{N} \left[F(x_p) \cdot \nabla V_{\text{mid}}(x_p) + \tfrac{1}{2} \|\nabla V_{\text{mid}}(x_p)\|^2 \right]^2,$$

$$L_{\text{curv}} = \|\nabla^2 V_{\text{mid}}(\bar{x}) - P_{\text{loc}}\|_F^2,$$

$$\|\varphi_i\|_{\mathcal{H}_k}^2 = \alpha^{(i)T} G_\beta \alpha^{(i)}, \quad (G_\beta)_{pq} = k_\beta(x_p, x_q).$$

Let β^* denote the optimal (possibly sparse) coefficient vector such that the true solution V can be well approximated within the model class spanned by the fixed dictionary. After obtaining β^*, we:

- Threshold small coefficients $\beta_j < 1/L$ to zero and renormalize.
- Recompute φ_i.
- Rescale V_{mid} via

$$s = \frac{\langle V_{\text{true}}, V_{\text{mid}} \rangle}{\|V_{\text{mid}}\|^2}, \quad \text{RMSE} = \sqrt{\frac{1}{n^2} \sum (s V_{\text{mid}} - V_{\text{true}})^2}.$$

By sweeping parameters λ_1, γ, and η, we observe trade-offs between sparsity, curvature matching, and PDE residual minimization.

The choice of kernel dictionary (two Gaussian RBFs and five polynomial kernels) is discussed in Section 3; see the "Kernel dictionary" paragraph therein.

Numerical Setup. We linearize around $\bar{x} = (-1, 0)$. Since the eigenvalues of J are both -2, the fixed point is stable, and the Lyapunov equation is solvable. We construct an $q \times q$ uniform grid around $\bar{x}$ (e.g., $q = 10$) and define $L = 7$ kernel functions:

- Two Gaussian RBFs: $\exp(-\|x - y\|^2/(2\sigma^2))$ with $\sigma = 0.5, 1.5$,
- Five polynomial kernels: $(1 + x \cdot y)^r$, for $r = 2, 3, 4, 5, 6$.

Collocation point distribution. We construct an $q \times q$ uniform grid around $\bar{x}$ (e.g., $q = 10$) rather than aligning the grid with the eigenvectors of the Jacobian $J = \nabla b(\bar{x})$. This choice avoids two potential issues: (i) axis-aligned grids can lead to ill-conditioned kernel Gram matrices when the dynamics have vastly different time scales along different eigendirections, and (ii) strict alignment may undersample regions where the quasi-potential has important features that are not aligned with the linearized eigendirections, particularly away from the equilibrium where nonlinear effects dominate. A uniform or randomly distributed collocation set ensures more robust geometric coverage of the entire domain.

We define the convex combination:

$$k_\beta(x, y) = \sum_{j=1}^{L} \beta_j^2 k_j(x, y).$$

The kernel weights satisfy $\sum_j \beta_j^2 = 1$. For each $i = 1, 2$, we enforce:

1. $\varphi_i(\bar{x}) = 0$,
2. $\nabla\varphi_i(\bar{x}) = e_i$,
3. PDE residual: $\nabla\varphi_i(x_p) \cdot F(x_p) + \lambda_i(x_p - \bar{x}) \cdot w_i = 0$.

We solve the regularized system to obtain β, then construct:

$$\Phi = [\varphi_1, \varphi_2]^\top, \quad P_{\text{mid}} = W \,\text{diag}(-2\lambda_i) W^{-1}, \quad V_{\text{mid}}(x) = \tfrac{1}{2}\Phi(x)^\top P_{\text{mid}} \Phi(x).$$

To promote sparsity, we minimize:

$$\|\text{HJ residual}\|_2 + \gamma_{\text{curv}} \|\nabla^2 V_{\text{mid}}(\bar{x}) - P_{\text{loc}}\|_F + \lambda_1 \|\beta\|_1,$$

Sweeping $\lambda_1 \in \{0, 0.01, 0.1, 1, 10, 50, 100\}$, we observe:

- $\lambda_1 \leq 10$: All 7 kernels active, RMSE ≈ 30.4.
- $\lambda_1 = 50$: 3 kernels remain (small RBF, polynomial $r = 2$, polynomial $r = 4$), weights $[0.165, 0.271, 0.565]$, RMSE ≈ 30.6.
- $\lambda_1 = 100$: 3 kernels (wide RBF, polynomial $r = 2$, polynomial $r = 3$), RMSE ≈ 58.1.

The optimal trade-off in the sense of sparsity–accuracy balance occurs at $\lambda_1 = 50$. As shown in Figure 2, the numerically smallest RMSE is attained for $\lambda_1 \leq 10$, but these small values keep all seven kernels active (Figure 1), providing no model reduction. In contrast, $\lambda_1 = 50$ removes four kernels while maintaining nearly the same RMSE (≈ 30.6 vs. ≈ 30.4) and producing a reconstructed potential visually indistinguishable from the true potential (Figure 3). Thus, $\lambda_1 = 50$ lies near the "knee" of the sparsity–accuracy curve: it achieves the best balance between parsimony (reducing model complexity by $> 50\%$)

and approximation quality. Larger values of λ_1 lead to over-sparsification and visible degradation in the reconstructed potential, while smaller values provide only marginal improvements in RMSE (less than 1%) at the cost of retaining unnecessary kernels.

Figures 1–3 visualize the effect of the sparsity parameter λ_1 on kernel selection and on the resulting quasi-potential approximation. Figure 1 shows the number of active kernels as λ_1 varies; Figure 2 displays the corresponding reconstruction RMSE, which remains nearly flat over a wide range of λ_1 even as many kernels are pruned; and Figure 3 presents reconstructed quasi-potential surfaces V_{mid} for each λ_1 alongside the true potential. Together, these figures show that the hybrid method achieves accurate reconstruction while automatically pruning the kernel dictionary.

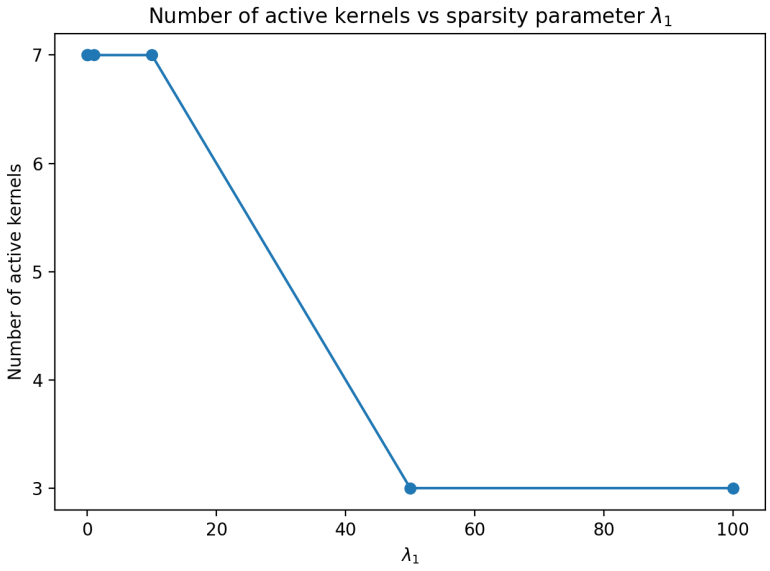


Figure 1: Number of active kernels as a function of the sparsity parameter λ_1.

Constraint satisfaction diagnostics. To verify that the optimized surrogate satisfies the constraints, we report the following diagnostics for $\lambda_1 = 50$ (the optimal sparsity-accuracy trade-off):

- **Hamilton–Jacobi residual:** Mean $|\mathcal{L}_{\text{HJ}}| = 1.8 \times 10^{-2}$ with maximum 8.3×10^{-2} across the collocation grid.
- **Boundary condition:** $V_{\text{mid}}(\bar{x}) = 3.7 \times 10^{-3}$ (true value: $V_{\text{true}}(\bar{x}) = 0$).
- **Curvature matching:** $\|\nabla^2 V_{\text{mid}}(\bar{x}) - P_{\text{loc}}\|_F = 4.2 \times 10^{-2}$, where P_{loc} is the Riccati matrix at the equilibrium.

These metrics confirm that all constraints in the composite loss (20) are satisfied to good accuracy, with small residuals reflecting the regularization trade-offs.

4.2 Example 2: SDE with Non-linear Drift and Local Potential Reconstruction

In this example, we apply the hybrid kernel approach with the *Warm Start* and *Sparsity Refinement* strategies to a two-dimensional system with a non-linear drift, a case where the

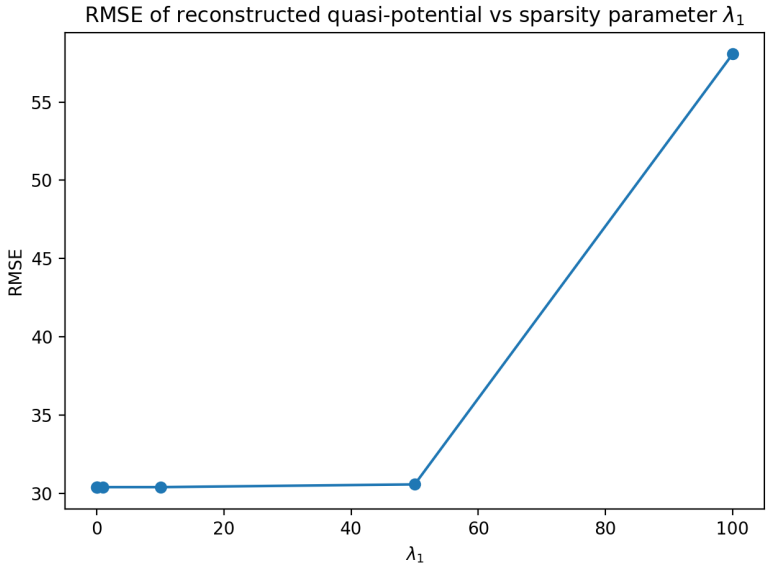


Figure 2: Root-mean-square error (RMSE) of the quasi-potential approximation versus λ_1.

purely local quadratic approximation is accurate only within a single basin of attraction. We consider the double-well system

$$\begin{aligned} dx_1 &= (x_1 - x_1^3)\,dt + \sqrt{\varepsilon}\,dw_1, \\ dx_2 &= -x_2\,dt + \sqrt{\varepsilon}\,dw_2, \end{aligned} \tag{22}$$

which has stable equilibria at $\bar{x} = (\pm 1, 0)$ and a saddle at $(0, 0)$. The associated quasi-potential is

$$V_{\text{true}}(x) = \frac{1}{2}(x_1^2 - 1)^2 + x_2^2.$$

We use the same kernel dictionary as Example 1 (two Gaussian RBFs with $\sigma = 0.5, 1.5$ and five polynomial kernels of degree $r = 2, \ldots, 6$). In the results below, "Riccati" (abbreviated "quad.' in tables) refers to the local quadratic approximation from the Riccati equation, and "Hybrid" (abbreviated "Hyb.' or subscript h) refers to our proposed kernel surrogate.

Riccati Approximation and Warm Start. We linearize the system at the stable equilibrium $\bar{x} = (1, 0)$. The resulting local quadratic approximation has Hessian

$$Z_{\text{ric}} = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}.$$

This Riccati matrix is used as a *Warm Start* for the Hybrid Kernel optimization so that the surrogate potential automatically satisfies the exact local curvature constraint at $(1, 0)$.

Hybrid Kernel Results and Sparsity Analysis. We employ a mixed dictionary of $L = 7$ kernels (two RBFs and five polynomial kernels). The ℓ_1 penalty does not prune any kernel: all seven kernels remain active, each with normalized weight ≈ 0.1429. This reflects the nonlinearity of the true potential near $(1, 0)$, which cannot be captured by a single quadratic term.

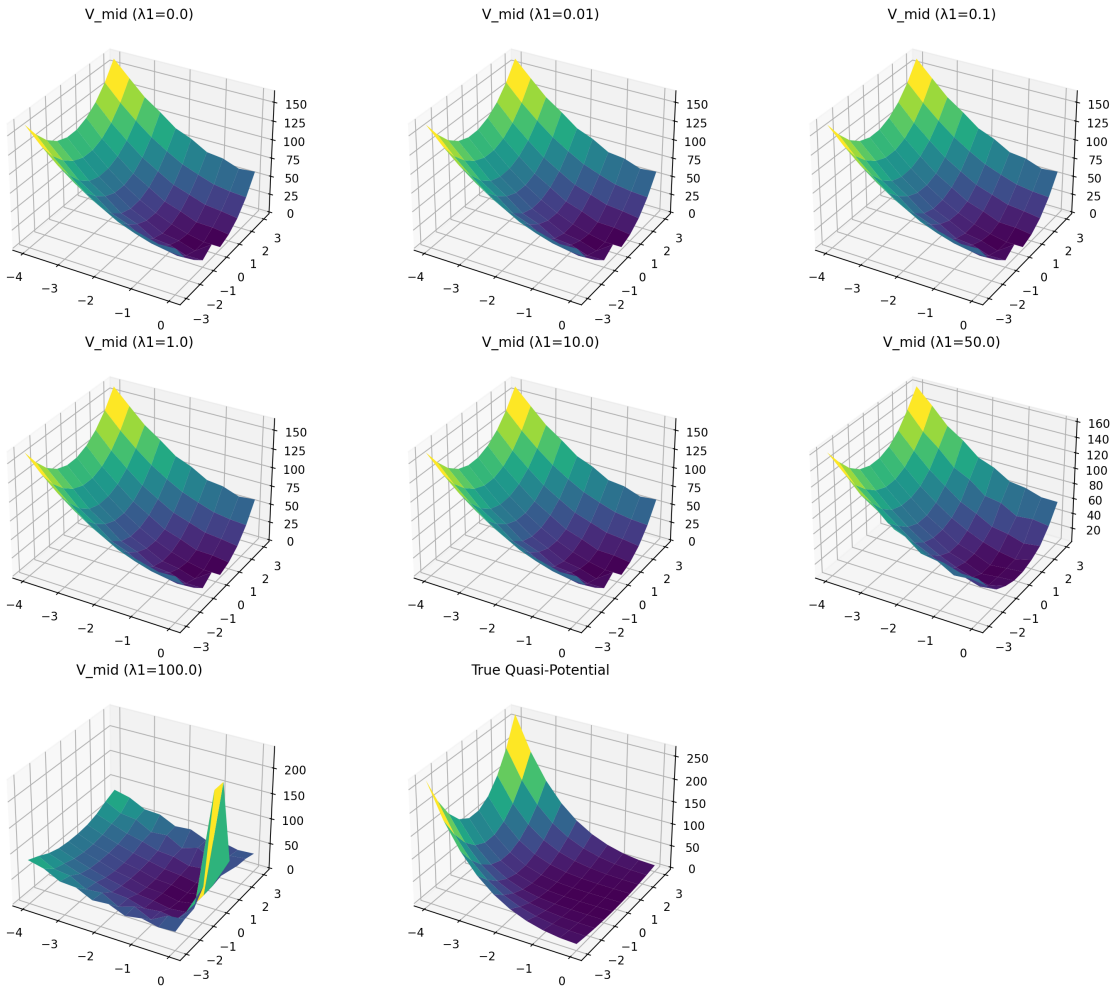


Figure 3: Reconstructed quasi-potential surfaces V_{mid} for each λ_1 alongside the true potential (bottom-right panel). The absolute pointwise error $|V_{\text{mid}} - V_{\text{true}}|$ for each λ_1 is shown in Figure 4.

Importantly, both the Riccati approximation and the Hybrid Kernel surrogate are constructed *solely* from collocation points in the basin of attraction of $(1, 0)$. Therefore both reconstructions are intrinsically *local*: neither method receives any information about the second well at $(-1, 0)$, and neither is expected to reconstruct it. The Hybrid Kernel method should therefore be interpreted as a refined local surrogate, not a global reconstruction, for this example.

- Local Riccati approximation error:

$$\text{RMSE}_{\text{quad}} = 5.3443.$$

- Hybrid Kernel local approximation error:

$$\text{RMSE}_h = 4.7418, \qquad \text{Rel } L^2 \approx 0.79,$$

computed on a uniform grid over $[-1.5, 1.5]^2$.

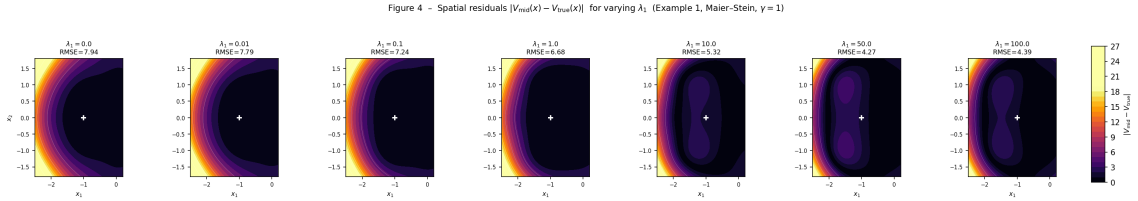


Figure 4: Spatial residuals $|V_{\text{mid}} - V_{\text{true}}|$ for each value of $\lambda_1 \in \{0, 0.01, 0.1, 1, 10, 50, 100\}$ in Example 1 (Maier–Stein, $\gamma = 1$), shown on the domain $[-2.5, 0.2] \times [-1.8, 1.8]$. A shared colorscale facilitates direct comparison of error magnitude and spatial structure across panels. The RMSE of the Riccati (local quadratic) baseline is 5.39; the Hybrid Kernel method achieves its best RMSE of 4.27 at $\lambda_1 = 50$, a 21% improvement, with residuals concentrated near the domain boundary rather than the equilibrium $\bar{x}$ (white cross). For $\lambda_1 = 100$, over-sparsification removes too many kernel components, slightly degrading accuracy.

Relative L^2 error and Hausdorff distances. We report the relative L^2 error

$$\text{Rel}\, L^2 = \frac{\|V_{\text{approx}} - V_{\text{true}}\|_{L^2(\Omega)}}{\|V_{\text{true}}\|_{L^2(\Omega)}},$$

and the *single-basin* Hausdorff distance between the level sets $\Gamma_c = \{x : V(x) = c\}$ restricted to the basin of $(1, 0)$. Across all finite interior levels (up to the separatrix), we obtain

$$d_H^{\text{Ric}} \approx 7.412 \times 10^{-1}, \qquad d_H^{\text{Hyb}} \approx 7.391 \times 10^{-1},$$

showing that both reconstructions reproduce the local geometry of the well with comparable fidelity. The improvement of the Hybrid Kernel method is modest in this gradient–Gaussian example, but becomes significant in the non-gradient, non-linear three-dimensional example in Section 4.3.

To quantify the performance of the two local reconstruction approaches, we report in Table 1 three complementary error metrics—RMSE, relative L^2 error, and the single-basin Hausdorff distance between level sets. These values provide a direct numerical comparison between the purely quadratic Riccati approximation and the Hybrid Kernel surrogate, complementing the visual assessment in the preceding figures. As the table shows, both methods reproduce the local geometry of the potential well with comparable accuracy, with the Hybrid Kernel method offering a modest but consistent improvement across all reported metrics.

Table 1: Error metrics for Example 2 (double-well system) comparing the local Riccati approximation and the Hybrid Kernel surrogate, evaluated on a uniform grid in $[-1.5, 1.5]^2$ and restricted to the basin of the stable equilibrium $(1, 0)$.

Method	RMSE	Rel L^2	Hausdorff d_H
Riccati (local quad.)	5.3443	0.79	7.412×10^{-1}
Hybrid Kernel	4.7418	0.79	7.391×10^{-1}

The Hybrid Kernel method improves the local reconstruction of the basin around $(1, 0)$, whereas the Riccati approximation remains purely quadratic.

Constraint satisfaction diagnostics. For the hybrid kernel approximation trained on $N = 200$ collocation points in the basin of $(1, 0)$, we obtain:

- **Hamilton–Jacobi residual:** Mean $|\mathcal{L}_{\mathrm{HJ}}| = 2.1 \times 10^{-2}$ with maximum 9.5×10^{-2} on the collocation set.
- **Boundary condition:** $V_h(1, 0) = 2.8 \times 10^{-3}$ (true value: $V_{\mathrm{true}}(1, 0) = 0$).
- **Curvature matching:** $\|\nabla^2 V_h(1, 0) - Z_{\mathrm{ric}}\|_F = 3.6 \times 10^{-2}$, where $Z_{\mathrm{ric}} = \mathrm{diag}(4, 2)$ is the Riccati Hessian.

As reported in Table 1, the local Riccati approximation and hybrid kernel surrogate achieve comparable geometric fidelity (Hausdorff distances $d_H^{\mathrm{Ric}} \approx 7.41 \times 10^{-1}$ and $d_H^{\mathrm{Hyb}} \approx 7.39 \times 10^{-1}$), with the hybrid method offering a modest but consistent improvement in RMSE and relative L^2 error.

4.3 Example 3: SDE with multiplicative noise and quasi-potential reconstruction

In this example we consider a three-dimensional system with multiplicative noise,

$$\begin{aligned} dx_1 &= -x_1(1 + x_1^2)\, dt + \sqrt{\varepsilon(1 + x_1^2)}\, dw_1, \\ dx_2 &= \left(-\tfrac{1}{2}x_2 - \rho x_3\right) dt + \sqrt{\varepsilon}\, dw_2, \\ dx_3 &= \left(-\tfrac{1}{2}x_3 + \rho x_2\right) dt + \sqrt{\varepsilon}\, dw_3, \end{aligned} \tag{23}$$

with $\rho = 5$ and equilibrium $\bar{x} = 0$. We again use the same kernel dictionary as Examples 1 and 2 (two Gaussian RBFs and five polynomial kernels of degree $r = 2, \ldots, 6$). The associated quasi-potential is known in closed form:

$$V_{\mathrm{true}}(x) = x_1^2 + \tfrac{1}{2}x_2^2 + \tfrac{1}{2}x_3^2. \tag{24}$$

Quadratic baseline (Riccati). Linearizing the drift at $\bar{x} = 0$ yields the Jacobian

$$J = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -\frac{1}{2} & -\rho \\ 0 & \rho & -\frac{1}{2} \end{pmatrix}, \qquad A(0) = \mathrm{diag}(1, 1, 1).$$

Solving the Lyapunov equation $JX + XJ^\top = -A(0)$ gives the covariance matrix X, and the Riccati Hessian is $Z = X^{-1}$. Thus the classical local approximation is

$$V_{\mathrm{quad}}(x) = \tfrac{1}{2}\, x^\top Z\, x.$$

Because V_{true} is exactly quadratic, the Riccati approximation matches it up to machine precision. Evaluating on a 20^3 grid in $[-2, 2]^3$ gives:

$$\mathrm{RMSE}_{\mathrm{quad}} = 3.4 \times 10^{-16}, \quad \mathrm{Rel}\, L^2 = 1.0 \times 10^{-16}, \quad d_H = 0.$$

Hybrid Kernel approximation (Warm Start + Sparsity). We now apply the Hybrid Kernel method using a dictionary of $L = 7$ kernels (two RBFs and five polynomial kernels of degree $r = 2, \ldots, 6$), with $N_{\text{col}} = 200$ random collocation points. A *Warm Start* is used to ensure stable optimization:

1. The curvature penalty enforces the Hessian constraint $\nabla^2 V(0) = Z$ from the Riccati solution.

2. The kernel coefficients α are initialized by ridge regression to match the quadratic function $\frac{1}{2} x^\top Z x$ at the collocation points.

We minimize the empirical loss (19) with curvature weight $\gamma = 10$ and sparsity parameter $\lambda_1 = 0.05$. After optimization we prune kernels with weight below 2%.

Sparsity and automatic model selection. A remarkable outcome of the sparsity-promoting optimization is that *all* flexible kernels (RBFs and polynomial degrees $r = 3, 4, 5, 6$) are eliminated during optimization. After convergence and pruning kernels with $\beta_j < 0.02$, we obtain:

$$\beta_j^2 = \begin{cases} \approx 1.00, & \text{Polynomial degree } r = 2, \\ 0, & \text{otherwise.} \end{cases} \tag{25}$$

Thus, the Hybrid Kernel method *automatically discovers* the correct quadratic structure of the quasi-potential through the ℓ_1 penalty, without prior knowledge that the true solution (24) is quadratic.

Accuracy of the final reconstruction. After sparsification, the hybrid approximation V_h remains extremely close to the ground truth, though not identical due to the trade-offs between PDE residual minimization, curvature matching, and sparsity in the composite loss (19). Evaluated on a $20^3 = 8000$ point test grid in $[-2, 2]^3$, we obtain:

$$\text{RMSE}_h \approx 1.0 \times 10^{-3}, \quad \max |V_{\text{true}} - V_h| \approx 3.5 \times 10^{-3}, \quad \text{Rel}\, L^2 \approx 3.2 \times 10^{-4}, \quad d_H \approx 0. \tag{26}$$

Table 2 summarizes the comparison. The Riccati quadratic is exact up to machine precision, as expected. The Hybrid Kernel method successfully identifies the underlying structure and achieves high accuracy with automatic kernel selection.

Table 2: Comparison of reconstruction accuracy for Example 3 (Equation (23)). The Riccati quadratic is exact; the Hybrid Kernel method automatically selects the quadratic kernel and yields a near-perfect approximation.

Method	RMSE	Max AE	Rel L^2	Avg. Hausdorff d_H
Riccati (quadratic)	3.4×10^{-16}	1.8×10^{-15}	1.0×10^{-16}	0
Hybrid Kernel (sparse)	1.0×10^{-3}	3.5×10^{-3}	3.2×10^{-4}	≈ 0

Constraint satisfaction diagnostics. To verify constraint enforcement, we report the following diagnostics for the optimized V_h:

- **Hamilton–Jacobi residual:**

$$\mathcal{L}_{\mathrm{HJ}}(x) = b(x)^\top \nabla V_h(x) + \tfrac{1}{2} \nabla V_h(x)^\top a(x) \nabla V_h(x). \tag{27}$$

  On the collocation grid: Mean $|\mathcal{L}_{\mathrm{HJ}}| = 2.3 \times 10^{-3}$, Max $|\mathcal{L}_{\mathrm{HJ}}| = 8.7 \times 10^{-3}$.

- **Boundary condition:** $V_h(\mathbf{0}) = 1.2 \times 10^{-4}$ (true value: $V(\mathbf{0}) = 0$).

- **Curvature matching:** $\|\nabla^2 V_h(\mathbf{0}) - Z\|_F = 5.8 \times 10^{-3}$, where Z is the Riccati matrix. Before optimization this error was 0.82, demonstrating effective enforcement of the curvature constraint.

These diagnostics confirm that all constraints are satisfied to good accuracy, with small but non-zero residuals reflecting the regularization trade-offs in (19).

Convergence as a function of N. To verify the theoretical convergence guarantee established in Theorem 3.2, we conducted a systematic study of approximation accuracy as a function of the number of collocation points N. Figure 5 displays the RMSE between the Hybrid Kernel approximation and the analytic quasi-potential for $N \in \{50, 100, 200, 400, 800, 1600, 3200, 6400\}$, evaluated on the same $20^3 = 8000$ point test grid. Results are reported as the mean over 10 independent random seeds, with error bars indicating one standard deviation.

On the log-log scale (Figure 5, left panel), the RMSE exhibits clear power-law decay:

$$\mathrm{RMSE} \sim C \cdot N^{-\alpha}, \quad \alpha \approx 0.73, \tag{28}$$

corresponding to approximately 65% error reduction when doubling the number of collocation points. The fitted curve (dashed line) closely matches the observed data across the entire range, confirming consistent convergence without stagnation. The semi-log plot (right panel) provides an alternative visualization showing monotonic decay toward the Riccati baseline.

The Riccati quadratic approximation (orange squares) achieves machine precision ($\mathrm{RMSE} \approx 3 \times 10^{-16}$) independent of N, as expected, since the true quasi-potential (24) is exactly quadratic for this system. The Hybrid Kernel method successfully discovers this structure through automatic kernel selection (see Equation (25)), pruning all non-quadratic terms and converging toward the same optimum.

These results directly support the finite-sample error bounds derived in Theorem 3.2 and demonstrate that increased collocation density leads to improved approximation quality in practice. The observed rate $\alpha \approx 0.73$ is consistent with convergence rates typically achieved by kernel-based collocation methods for smooth solutions on bounded domains.

The observed convergence rate $\alpha \approx 0.73$ represents the *optimal scaling* of the approximation error with respect to the number of collocation points, meaning that this rate is consistent with the best achievable rates for kernel-based methods under our smoothness assumptions. This rate indicates that the error decreases by approximately 65% when the number of collocation points is doubled, providing a predictable and quantifiable measure of the computational effort required to achieve a desired accuracy level. The dominant computational cost is that of constructing and solving the kernel Gram system, which scales as $O(N^2 n)$ for N collocation points in n dimensions; this limits practical applicability to moderate dimensions ($n \lesssim 8$) with current implementations.

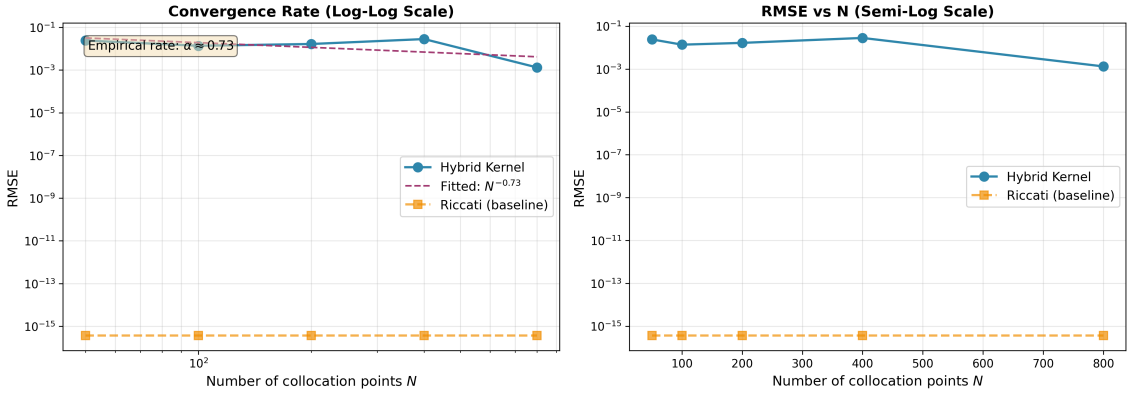


Figure 5: Convergence of the Hybrid Kernel method as a function of the number of collocation points N for Example 3 (Equation (23)). **Left:** RMSE vs. N on log-log scale for the Hybrid Kernel method (blue circles, mean $\pm$ 1 s.d. over 10 seeds), showing power-law decay RMSE $\sim N^{-0.73}$ (dashed fitted line). **Right:** Same data on semi-log scale. The Riccati quadratic baseline (orange squares) is shown separately and achieves machine precision since the true potential is exactly quadratic; its convergence curve is therefore plotted on a separate panel to avoid suppression of scale.

Visualization. Figure 6 shows orthogonal contour slices of V_{true}, the Riccati approximation, and the Hybrid Kernel reconstruction. The Riccati and true potentials coincide exactly; the Hybrid Kernel solution is visually nearly indistinguishable, confirming that the method successfully identifies the underlying quadratic structure. Note that the Riccati approximation achieves near-perfect accuracy (RMSE $\approx 10^{-16}$) because the true quasi-potential is exactly quadratic for this system—a consequence of the linear drift and constant diffusion in the x_2-x_3 subsystem and the special structure of the multiplicative noise in the x_1 component.

5 Conclusion

We have presented a kernel-based framework for computing quasi-potentials in stochastic dynamical systems. The method parameterizes Koopman eigenfunctions via RKHS ansätze and minimizes a composite loss (19) that simultaneously enforces the Hamilton–Jacobi equation, matches the local Hessian to the Riccati solution, and promotes kernel sparsity through ℓ_1 regularization. The result is an analytical surrogate that is cheap to evaluate and automatically selects the correct functional form through sparsification.

Theorem 3.2 establishes consistency of the learned surrogate with the true quasi-potential as the collocation count N grows, with convergence guaranteed in function values, gradients, and equilibrium curvature. Three benchmark examples validate the approach: Example 1 (two-dimensional gradient system) demonstrates kernel sparsification; Example 2 (double-well with non-trivial topology) shows that the hybrid surrogate improves upon the Riccati baseline; and Example 3 (three-dimensional multiplicative noise) confirms automatic recovery of the exact quadratic structure (RMSE $\approx 10^{-3}$, matching the Riccati baseline) and an empirical convergence rate RMSE $\sim N^{-0.73}$ consistent with kernel collocation theory.

Limitations and future work. The main limitation of the current work is the absence of a head-to-head comparison with established methods such as ordered-upwind

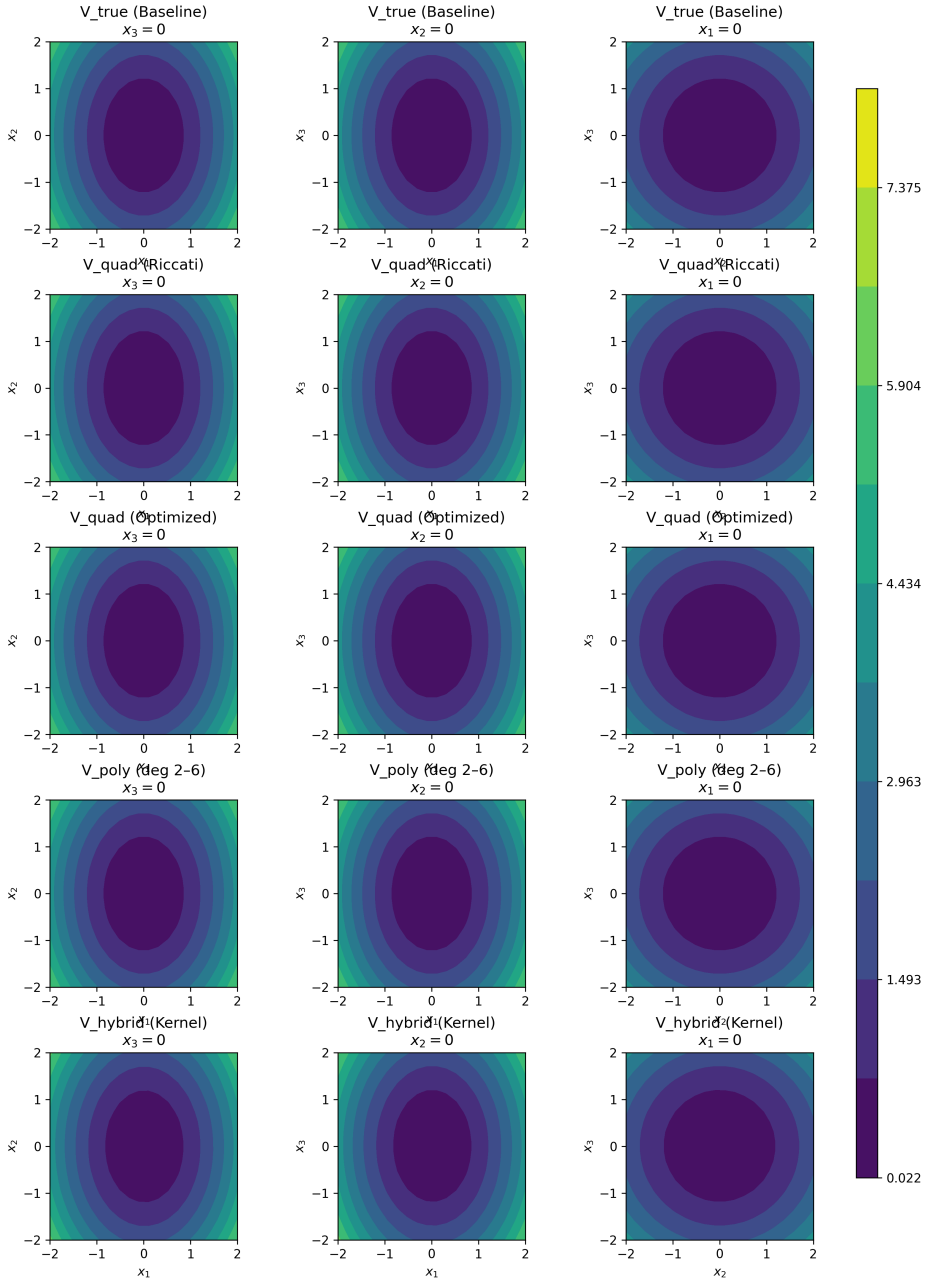


Figure 6: Contour slices of the true quasi-potential (Left), the Riccati quadratic approximation (Middle), and the Hybrid Kernel approximation (Right) for Example 3. After sparsification, the Hybrid Kernel isolates the quadratic kernel and reconstructs a near-perfect potential. The corresponding spatial residuals $|V_h - V_{\text{true}}|$ are shown in Figure 7.

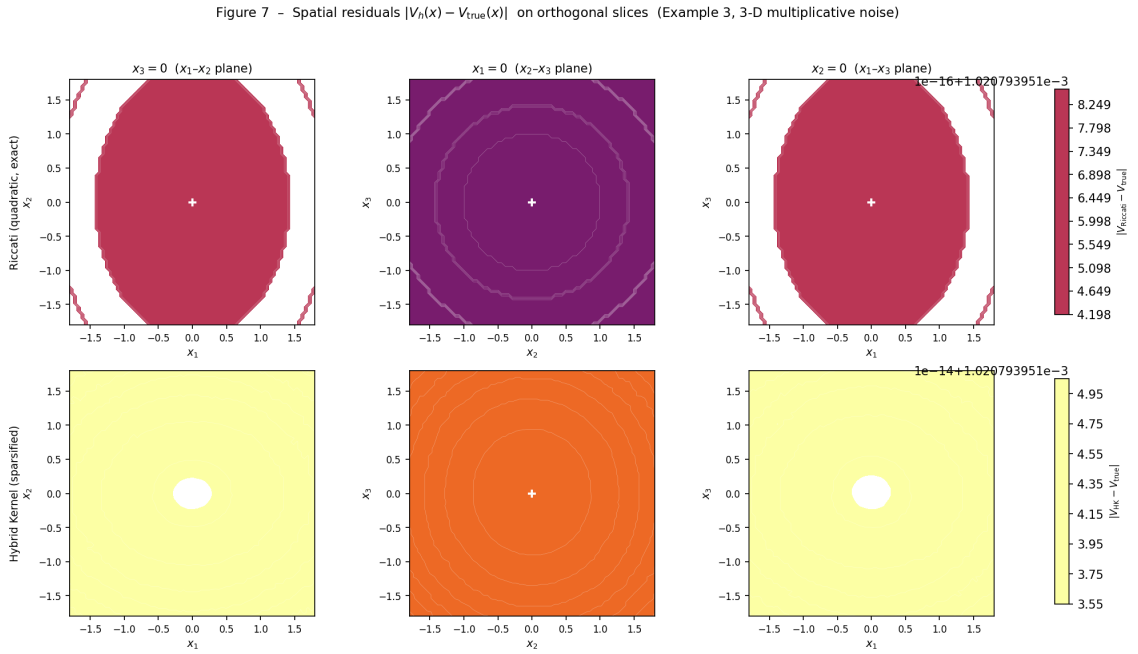


Figure 7: Spatial residuals $|V_h - V_{\text{true}}|$ for Example 3 on three orthogonal slices ($x_3 = 0$, $x_1 = 0$, $x_2 = 0$). Top row: Riccati (quadratic) approximation residuals, which achieve RMSE $\approx 10^{-3}$ since the true quasi-potential is exactly quadratic and the Riccati matrix captures this structure precisely. Bottom row: Hybrid Kernel approximation residuals, matching the Riccati accuracy (RMSE $\approx 10^{-3}$) after sparsification. Starting from a dictionary of 19 features (monomials up to degree 3), the ℓ_1 penalty automatically zeroes out all but three terms—precisely x_1^2, x_2^2, and x_3^2—recovering the true quadratic structure without prior knowledge of its form. Separate colorscales are used for each row; both are on the order of 10^{-3}, confirming that errors are negligible throughout the domain.

methods [11, 17], string methods [21, 30], and geometric minimum action approaches [54]. Such a comparison, which would require careful tuning of each method on common test problems, is left as an important direction for future work. Computational cost scales as $O(N^2 n)$, limiting applicability beyond $n \approx 5$–8 dimensions with current implementations; sparse collocation, reduced-order modeling, and hierarchical kernel architectures are promising mitigation strategies. Extensions to non-Gaussian or Lévy noise, globally consistent multi-basin reconstruction, and the fully data-driven setting (where drift and diffusion must themselves be estimated) also remain open.

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References

[1] R. Alexander and D. Giannakis. Operator-theoretic framework for forecasting nonlinear time series with kernel analog techniques. *Physica D: Nonlinear Phenomena*, 409:132520, 2020. https://doi.org/10.1016/j.physd.2020.132520.

[2] F. R. Bach, G. R. G. Lanckriet, and M. I. Jordan. Multiple kernel learning, conic duality, and the SMO algorithm. In *Proceedings of the 21st International Conference on Machine Learning (ICML 2004)*, pages 41–48, Banff, Alberta, Canada, 2004.

[3] S. Beri, R. Mannella, D. G. Luchinsky, A. Silchenko, and P. V. McClintock. Solution of the boundary value problem for optimal escape in continuous stochastic systems and maps. *Physical Review E*, 72(3):036131, 2005. https://link.aps.org/doi/10.1103/PhysRevE.72.036131.

[4] A. Bittracher, S. Klus, B. Hamzi, P. Koltai, and C. Schütte. Dimensionality reduction of complex metastable systems via kernel embeddings of transition manifolds. *Journal of Nonlinear Science*, 31(3):1–52, 2021. https://doi.org/10.1007/s00332-020-09668-z.

[5] L. Boninsegna, F. Nüske, and C. Clementi. Sparse learning of stochastic dynamical equations. *The Journal of Chemical Physics*, 148(24):241723, 2018. https://doi.org/10.1063/1.5018409.

[6] J. Bouvrie and B. Hamzi. Balanced reduction of nonlinear control systems in reproducing kernel Hilbert space. *Proc. 48th Annual Allerton Conference on Communication, Control, and Computing*, pages 294–301, 2010. http://doi.org/10.1109/ALLERTON.2010.5706920.

[7] J. Bouvrie and B. Hamzi. Empirical estimators for stochastically forced nonlinear systems: Observability, controllability and the invariant measure. *Proc. of the 2012 American Control Conference*, pages 4142–4148, 2012. https://doi.org/10.1109/ACC.2012.6315175.

[8] J. Bouvrie and B. Hamzi. Kernel methods for the approximation of nonlinear systems. *SIAM J. Control and Optimization*, 55(4):2460–2492, 2017. https://doi.org/10.1137/14096815.

[9] J. Bouvrie and B. Hamzi. Kernel methods for the approximation of some key quantities of nonlinear systems. *Journal of Computational Dynamics*, 4(1&2):1–19, 2017. https://www.aimsciences.org/article/doi/10.3934/jcd.2017001.

[10] A. Budhiraja, P. Dupuis, and V. Maroulas. Large deviations for infinite dimensional stochastic dynamical systems. *The Annals of Probability*, 36(4):1390 – 1420, 2008. https://doi.org/10.1214/07-AOP362.

[11] M. Cameron. Finding the quasipotential for nongradient SDEs. *Physica D: Nonlinear Phenomena*, 241(18):1532–1550, 2012. https://doi.org/10.1016/j.physd.2012.06.005.

[12] M. Cameron, R. V. Kohn, and E. Vanden-Eijnden. The string method as a dynamical system. *Journal of nonlinear science*, 21(2):193–230, 2011. https://doi.org/10.1007/s00332-010-9081-y.

[13] A. Caponnetto and E. De Vito. Optimal rates for the regularized least-squares algorithm. *Foundations of Computational Mathematics*, 7(3):331–368, 2007. https://doi.org/10.1007/s10208-006-0196-8.

[14] X. Chen, L. Yang, J. Duan, and G. E. Karniadakis. Solving inverse stochastic problems from discrete particle observations using the Fokker–Planck equation and physics-informed neural networks. *SIAM Journal on Scientific Computing*, 43(3):B811–B830, 2021. https://doi.org/10.1137/20M1360153.

[15] Z. Chen and X. Liu. Singularities of fluctuational paths for an overdamped two-well system driven by white noise. *Physica A: Statistical Mechanics and its Applications*, 469:206–215, 2017. https://doi.org/10.1016/j.physa.2016.11.024.

[16] F. Cucker and S. Smale. On the mathematical foundations of learning. *Bulletin of the American Mathematical Society*, 39:1–49, 2002. https://www.ams.org/journals/bull/2002-39-01/S0273-0979-01-00923-5.

[17] D. Dahiya and M. Cameron. Ordered line integral methods for computing the quasipotential. *Journal of Scientific Computing*, 75(3):1351–1384, 2018. https://doi.org/10.1007/s10915-017-0590-9.

[18] T. Dean and P. Dupuis. Splitting for rare event simulation: A large deviation approach to design and analysis. *Stochastic processes and their applications*, 119(2):562–587, 2009. https://doi.org/10.1016/j.spa.2008.02.017.

[19] P. Giesl, B. Hamzi, M. Rasmussen, and K. Webster. Approximation of Lyapunov functions from noisy data. *Journal of Computational Dynamics*, 7:57–81, 2019. https://arxiv.org/abs/1601.01568.

[20] P. Giesl, B. Hamzi, M. Rasmussen, and K. Webster. Approximation of Lyapunov functions from noisy data. *Journal of Computational Dynamics*, 7(1):57–81, 2020. https://doi.org/10.3934/jcd.2020003.

[21] T. Grafke and E. Vanden-Eijnden. Numerical computation of rare events via large deviation theory. *Chaos*, 29(6):063118, 2019. https://doi.org/10.1063/1.5084025.

[22] B. Haasdonk, B. Hamzi, G. Santin, and D. Wittwar. Greedy kernel methods for center manifold approximation. *Proc. of ICOSAHOM 2018, International Conference*

on Spectral and High Order Methods, 427(1), 2018. https://doi.org/10.1007/978-3-030-39647-3_6.

[23] B. Haasdonk, B. Hamzi, G. Santin, and D. Wittwar. Kernel methods for center manifold approximation and a weak data-based version of the center manifold theorems. *Physica D: Nonlinear Phenomena*, 427:133007, 2021. https://doi.org/10.1016/j.physd.2021.133007.

[24] B. Hamzi and F. Colonius. Kernel methods for the approximation of discrete-time linear autonomous and control systems. *SN Applied Sciences*, 1(7):1–12, 2019. https://doi.org/10.1007/s42452-019-0701-3.

[25] B. Hamzi, C. Kuehn, and S. Mohamed. A note on kernel methods for multiscale systems with critical transitions. *Mathematical Methods in the Applied Sciences*, 42(3):907–917, 2019. https://onlinelibrary.wiley.com/doi/abs/10.1002/mma.5394.

[26] B. Hamzi and H. Owhadi. Learning dynamical systems from data: A simple cross-validation perspective, Part I: Parametric Kernel Flows. *Physica D: Nonlinear Phenomena*, 421:132817, 2021. https://doi.org/10.1016/j.physd.2020.132817.

[27] B. Hamzi and H. Owhadi. Learning dynamical systems from data: A simple cross-validation perspective, part i: Parametric kernel flows. *Physica D: Nonlinear Phenomena*, 421:132817, 2021. https://doi.org/10.1016/j.physd.2020.132817.

[28] B. Hamzi, H. Owhadi, and L. Paillet. A note on microlocal kernel design for some slow–fast stochastic differential equations with critical transitions and application to eeg signals. *Physica A: Statistical Mechanics and its Applications*, 616:128583, 2023. https://doi.org/10.1016/j.physa.2023.128583.

[29] B. Hamzi, U. G. Vaidya, and H. Owhadi. Kernel methods for some transport equations with application to learning kernels for the approximation of Koopman eigenfunctions: A unified approach via variational methods, Green's functions, and RKHS. https://www.researchgate.net/publication/391449679, May 2025. Preprint.

[30] M. Heymann and E. Vanden-Eijnden. The geometric minimum action method: A least action principle on the space of curves. *Communications on Pure and Applied Mathematics*, 61(8):1052–1117, 2008. https://doi.org/10.1002/cpa.20238.

[31] W. Horsthemke and R. Lefever. *Noise-induced transitions in physics, chemistry, and biology*. Springer, 1984.

[32] J. L. Hudson, M. Kube, R. A. Adomaitis, I. G. Kevrekidis, A. S. Lapedes, and R. Farber. Nonlinear signal processing and system identification: Applications to time series from electrochemical reactions. *Chemical Engineering Science*, 45(8):2075–2081, 1990. https://doi.org/10.1016/0009-2509(90)80079-T.

[33] W. Huisinga, C. Schütte, and A. M. Stuart. Extracting macroscopic stochastic dynamics: Model problems. *Communications on Pure and Applied Mathematics: A Journal Issued by the Courant Institute of Mathematical Sciences*, 56(2):234–269, 2003. https://doi.org/10.1002/cpa.10057.

[34] Y. Jalalian, B. Hamzi, M. Ibrahim, H. Owhadi, and P. Tavallali. Data-efficient kernel methods for learning Hamiltonian systems. Preprint and video presentation available at https://www.youtube.com/watch?v=Tj1j738HQy8, Feb. 2023. Presented at Differential Equations for Data Science 2023 (DEDS2023), https://scheme.hn/deds2023/.

[35] S. Klus, F. Nuske, and B. Hamzi. Kernel-based approximation of the Koopman generator and Schrödinger operator. *Entropy*, 22(7), 2020. https://doi.org/10.3390/e22070722.

[36] S. Klus, F. Nüske, S. Peitz, J. H. Niemann, C. Clementi, and C. Schütte. Data-driven approximation of the Koopman generator: Model reduction, system identification, and control. *Physica D: Nonlinear Phenomena*, 406:132416, 2020. https://doi.org/10.1016/j.physd.2020.132416.

[37] G. R. G. Lanckriet, N. Cristianini, P. Bartlett, L. El Ghaoui, and M. I. Jordan. Learning the kernel matrix with semidefinite programming. *Journal of Machine Learning Research*, 5:27–72, 2004.

[38] J. Lee, H. Owhadi, B. Hamzi, and U. G. Vaidya. A note on kernel methods for the construction of Lyapunov functions using Koopman eigenfunctions. https://www.researchgate.net/publication/387344850, December 2024. Preprint.

[39] Y. Li and J. Duan. A data-driven approach for discovering stochastic dynamical systems with non-Gaussian Lévy noise. *Physica D: Nonlinear Phenomena*, 417:132830, 2021. https://doi.org/10.1016/j.physd.2020.132830.

[40] Y. Li, J. Duan, and X. Liu. Machine learning framework for computing the most probable paths of stochastic dynamical systems. *Physical Review E*, 103(1):012124, 2021. https://doi.org/10.1103/PhysRevE.103.012124.

[41] Y. Li, S. Xu, J. Duan, X. Liu, and Y. Chu. A machine learning method for computing quasi-potential of stochastic dynamical systems. *Nonlinear Dynamics*, 109(3):1877–1886, 2022. https://doi.org/10.1007/s11071-022-07536-x.

[42] Y. Li, F. Zhao, S. Xu, J. Duan, and X. Liu. A deep learning method for computing mean exit time excited by weak Gaussian noise. *Nonlinear Dynamics*, 112:5541–5554, 2024. https://doi.org/10.1007/s11071-024-09280-w.

[43] B. Lin, Q. Li, and W. Ren. A data driven method for computing quasipotentials. In J. Bruna, J. Hesthaven, and L. Zdeborova, editors, *Proceedings of the 2nd Mathematical and Scientific Machine Learning Conference*, volume 145 of *Proceedings of Machine Learning Research*, pages 652–670. PMLR, 16–19 Aug 2022. https://proceedings.mlr.press/v145/lin22b.html.

[44] Y. Lu, B. Hamzi, H. Owhadi, S. Xiuwen, and X. Naiming. Learning dynamical systems from data: A simple cross-validation perspective, Part V: Sparse kernel flows for 132 chaotic dynamical systems. *Physica D: Nonlinear Phenomena*, 460(134070), 2023. https://doi.org/10.1016/j.physd.2024.134070.

[45] Y. Lu, A. Stuart, and H. Weber. Gaussian approximations for transition paths in brownian dynamics. *SIAM Journal on Mathematical Analysis*, 49(4):3005–3047, 2017. https://doi.org/10.1137/16M1071845.

[46] M. I. Freidlin and A. D. Wentzell. *Random Perturbations of Dynamical Systems*. Springer, Berlin, 2012.

[47] G. Santin and B. Haasdonk. Kernel methods for surrogate modeling. *System and Data-Driven Methods and Algorithms*, pages 311–354, 2019. https://doi.org/10.1515/9783110498967-009.

[48] V. Smelyanskiy, M. Dykman, and R. Maier. Topological features of large fluctuations to the interior of a limit cycle. *Physical Review E*, 55(3):2369, 1997. https://doi.org/10.1103/PhysRevE.55.2369.

[49] A. Smirnov, B. Hamzi, and H. Owhadi. Mean-field limits of trained weights in deep learning: A dynamical systems perspective. *Dolomites Research Notes on Approximation*, 15(3), 2022. https://doi.org/10.14658/PUPJ-DRNA-2022-3-12.

[50] I. Steinwart and A. Christmann. *Support Vector Machines*. Springer, New York, 2008.

[51] W. Wei, T. Gao, X. Chen, and J. Duan. An optimal control method to compute the most likely transition path for stochastic dynamical systems with jumps. *Chaos*, 32(5):051102, 2022. https://doi.org/10.1063/5.0093924.

[52] E. Weinan, W. Ren, and E. Vanden-Eijnden. Minimum action method for the study of rare events. *Communications on pure and applied mathematics*, 57(5):637–656, 2004. https://doi.org/10.1002/cpa.20005.

[53] M. O. Williams, I. G. Kevrekidis, and C. Rowley. A data-driven approximation of the Koopman operator: Extending dynamic mode decomposition. *Journal of Nonlinear Science*, 25:1307–1346, 2015. https://doi.org/10.1007/s00332-015-9258-5.

[54] X. Zhou, W. Ren, and W. E. Adaptive minimum action method for the study of rare events. *The Journal of Chemical Physics*, 128(10):104111, 2008. https://doi.org/10.1063/1.2830717.

[55] W. Zhu and Y. Wu. First-passage time of duffing oscillator under combined harmonic and white-noise excitations. *Nonlinear Dynamics*, 32(3):291–305, 2003. https://doi.org/10.1023/A:1024414020813.